Malhar Mahajan

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EDUCATION

Virginia Tech, Blacksburg, VA — B.S. Aerospace Engineering **May 2027**
Guidance, Control & Navigation Concentration — GPA 3.85/4.00

SKILLS

Languages: C/C++, Python, MATLAB
Interests: Simulink (code generation), closed-loop simulation, multi-rate systems, estimation/filtering, controller tuning/validation
Tools: Git/GitHub, Linux, ROS 2, MAVSDK, ArduPilot/PX4, SITL/HITL, Siemens NX, Creo, SolidWorks

EXPERIENCE

Embedded Controls Co-op, Caterpillar Inc. (Technology Division) — **August 2025 – Present**

- Architected situational speed target generators and wheel-level control logic to allow for scenario specific traction control.
- Developed multi-rate integration approach for vehicle sensing (IMU + wheel speed sensors with different sampling rates) targeting Caterpillar ECM/ECU deployment; generated embedded C code from Simulink for early on-ECM timing/sampling-rate feasibility testing.
- Validated and tuned traction speed estimation/control using Dynasty plant testing; reduced oscillations observed in real data by accounting for motor/actuator physical constraints, and tuned to minimize overshoot and false-trigger events.

AVA Lab (NAViTi) — **January 2026 – Present**

- Developing a UWB-based inter-vehicle cross-link for collaborative navigation using Qorvo/Decawave modules; configuring anchor/tag networks and collecting range/pose data for analysis.
- Built MATLAB tooling to log and parse UWB outputs (tag positions and/or ranging streams) and quantify anomalies (bias, dropouts, and outliers) under different geometries and LOS/NLOS conditions.
- Next steps: extract and validate raw range measurements, then fuse UWB + inertial/vehicle motion models (EKF/particle filter) to enable cooperative localization/SLAM and characterize accuracy vs. anchor layout and dynamics.

Advanced Controls Systems Laboratory — **August 2024 – August 2025**

- Developed a three-DOF thrust-stand testbed for small UAVs to support autonomous controller tuning; implemented a Genetic Algorithm (GA) workflow in MATLAB for parameter search and evaluation.
- Integrated Pixhawk 6C with an Odroid M1S running ROS 2 Galactic; built ROS 2 nodes in C++ for Dynamixel-actuated moment cancellation using rigid-body inertia/moment models.
- Implemented IMU signal processing (high-pass filtering) to derive acceleration from published position/velocity topics and validated against logged data.

Team Lead, Avionics and Programming, GoAERO @VTech — **August 2024 – September 2025**

- Led avionics/software integration for a GoAERO UAV; defined ROS 2 messaging, interfaces, and integration plan between flight stack, sensors, and onboard compute.
- Built simulation-to-flight validation workflow (SITL/HITL + telemetry/command pipeline) to de-risk integration and support flight-test readiness across subteams.
- Contributed to NASA University Innovation Prize ($30K) as part of a multidisciplinary team.

Simulations Intern, Caterpillar Inc. (GTMS) — **May 2025 – August 2025**

- Supported GTMS design decisions with thermal/CFD analyses and helped accelerate adoption of NX Nastran workflows across the group.

PROJECTS

Autonomous UAV — **May 2024 – August 2024**

- Built an autonomous quadrotor using a Jetson Nano and Intel RealSense D455; implemented real-time depth-based mapping/obstacle avoidance and target tracking in Python/OpenCV.
- Integrated MAVSDK flight-control interfaces; validated autonomy in JMavSim and SITL/HITL prior to flight testing.